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FFKAN: A Lightweight Hyperspectral Image Classification Network via Frequency Domain Decomposition and Physically Constrained Fourier KAN

Dongtao Mo, Xiaoqing Wan, *Member, IEEE*, Genping Zhao, Huilu Tang, Zhize Li

***Abstract*—Hyperspectral image (HSI) classification requires effective modeling of complex spectral–spatial variations while suppressing spectral redundancy and maintaining computational efficiency. Existing frequency-domain methods often use fixed transformations, while KAN-based approaches typically serve as generic nonlinear modules, leaving HSI-specific frequency characteristics insufficiently explored. To address these limitations, this paper proposes FFKAN, a lightweight frequency-aware network that integrates learnable three-dimensional wavelet decomposition with a physically constrained Fourier Kolmogorov–Arnold representation. First, a learnable 3D wavelet frequency decomposition module (L3D-WFDM) decomposes the input into multiple spectral–spatial subbands, allowing low-frequency structural information and high-frequency discriminative details to be jointly captured. The resulting subbands are aggregated into compact representations and enhanced by 3D positional encoding. Subsequently, a physically constrained Fourier KAN module (PC-FKAN) combines residual 3D spatial context enhancement with learnable Fourier basis functions to efficiently model nonlinear spectral–spatial dependencies. Experiments on the Pavia University, Salinas, Xuzhou, and WHU-Hi-HanChuan datasets demonstrate that FFKAN achieves competitive classification performance with fewer parameters and lower computational complexity than several state-of-the-art methods, validating its effectiveness and lightweight design.**

***Index Terms*—Hyperspectral Image Classification, Frequency-Domain Decomposition, Fourier Kolmogorov–Arnold Network, Lightweight.**

I. Introduction

Hyperspectral images (HSIs) capture hundreds of contiguous spectral bands for each pixel, providing detailed spectral signatures beyond conventional RGB imagery [1]. Owing to their rich spectral–spatial information, HSI classification has been widely applied in diverse fields, ranging from ecological surveillance [2] and precision farming [3] to mineral prospecting [4], industrial damage detection [5], medical diagnostics [6], food safety [7], and cultural heritage preservation [8]. However, the high dimensionality of hyperspectral data also brings several challenges, including limited labeled samples, strong spectral redundancy, high computational complexity, and the Hughes phenomenon [9]. Therefore, effective spectral–spatial feature learning remains a fundamental issue in HSI classification.

In recent years, deep learning [10] has enabled significant progress in accurate HSI classification. Representative models, including stacked autoencoder (SAE) [11], deep belief networks (DBNs) [12], recurrent neural networks (RNNs) [13], graph neural networks (GNNs) [14], and convolutional neural networks (CNNs) [15], have been introduced to automatically learn discriminative representations from hyperspectral data. These methods laid a foundation for subsequent deep-learning-based HSI classification research. Among them, CNN-based models have become widely adopted owing to their strong capability in extracting local spatial features.

CNNs learn hierarchical spectral–spatial representations by using local convolutional kernels to capture neighboring pixel correlations and transform low-level features into high-level semantics [16]. To improve performance with limited labeled samples, Yao et al. [17] proposed semi-active CNNs (SA-CNNs), which combine active learning, pseudolabel generation, and superpixel segmentation to select reliable samples for iterative training. Zhou et al. [18] introduced quaternion CNNs and designed SQNet to map real HSI features into quaternion representations, enhancing structural feature modeling under small-sample conditions. Zhang et al. [19] proposed NESSG-GCN, which integrates gated graph convolution and pixel-level CNNs to capture both non-Euclidean and Euclidean spectral–spatial features while alleviating the limited receptive field of conventional convolutions.

To overcome the limitations of local convolutional modeling, Transformers use self-attention to capture long-range dependencies and learn high-level semantic representations beyond the receptive field of conventional convolutions [20]. Sun et al. [21] proposed the spectral–spatial feature tokenization transformer (SSFTT), which combines 3-D and 2-D convolutions with a Gaussian weighted tokenizer to convert shallow spectral–spatial features into tokens for efficient semantic

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Dongtao Mo, Xiaoqing Wan, Huilu Tang, and Zhize Li are with the Hengyang Normal University, College of Computer Science and Technology, Hengyang, 421002, China (e-mail: modongtao_hysf@163.com; e-mail: wanxiaoqingkaixin@163.com; e-mail: tanghuilu_hysf@163.com; e-mail: lizhize_hysf@163.com;).

Xiaoqing Wan is with the Hunan Provincial Key Laboratory of Intelligent Information Processing and Application, Hengyang, 421002, China (e-mail: wanxiaoqingkaixin@163.com).

Genping Zhao is with the School of Computer Science and Technology, Guangdong University of Technology, Guangzhou, 510006, China (e-mail: genping.zhao@gdut.edu.cn).

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learning. Roy et al. [22] introduced morphFormer, which incorporates learnable spectral and spatial morphological operations into the attention framework to enhance structural, shape, and token-level interactions. Zhang et al. [23] proposed the tensor transformer (TT), which directly processes raw HSI tensor data and employs tensor self-attention to capture long-range spatial–spectral structures while avoiding structural disruption caused by sequentialization.

More recently, Mamba models based on selective state-space models [24] have attracted increasing attention for their ability to model long-range dependencies with linear computational complexity. He et al. [25] proposed IGroupSS-Mamba, which partitions high-dimensional HSI features into nonoverlapping interval groups and performs multidirectional S6-based scanning to extract nonredundant global spatial–spectral information with reduced computational cost. Ahmad et al. [26] introduced MorpMamba, which integrates morphological operations into the Mamba framework to generate and enhance spatial–spectral tokens, improving structural and shape feature representation with high parameter efficiency. He et al. [27] further proposed HSI-MFormer, a hybrid Mamba–Transformer framework that combines inner-scale transformer experts and cross-scale Mamba experts to jointly model short-range dependencies and long-range multiscale spatial–spectral interactions.

Beyond architectural advances, recent studies have explored hyperspectral band selection to reduce spectral redundancy and improve generalization. Li et al. [28] proposed a multi-teacher meta-learning framework for zero-shot heterogeneous band selection. Feng et al. [29] further developed an uncertainty-aware meta-learning approach for few-shot heterogeneous scenarios. Chen et al. [30] introduced a heterogeneous graph convolutional self-representation network to model pixel- and band-level relationships. These studies highlight the importance of redundancy reduction, structural modeling, and generalization for effective HSI analysis.

In addition to CNN-, Transformer-, and Mamba-based architectures, Kolmogorov–Arnold networks (KANs) [31] have recently offered a new perspective for nonlinear feature modeling. By replacing the fixed activation functions in conventional multilayer perceptron (MLP) [32] with learnable edge-wise functions, KANs provide a more flexible representation learning mechanism. Han et al. [33] proposed KANEFT, which constructs linear and convolutional structures with KAN and integrates multi-scale fusion, spatial attention, and cross-attention to extract irregular local features and complementary information. Feng et al. [34] introduced HPFN, a fast KAN-based hierarchical progressive fusion network that explores HSI features at pixel, patch, and global levels while reducing computational cost and redundancy. Zhang et al. [35] proposed HyperFKAN, which replaces B-spline functions with Fourier series and introduces a periodic spatial shift module to improve spectral–spatial nonlinear feature learning and computational efficiency.

Despite the remarkable progress of CNN, Transformer, Mamba, and KAN-based methods, several challenges remain. CNNs effectively extract local spectral–spatial features, but fixed convolutional kernels and limited receptive fields restrict their ability to model long-range dependencies and continuous spectral variations. Transformers and Mamba models improve global dependency modeling, yet tokenization or sequential scanning may disturb the inherent spectral–spatial structure of HSIs and increase computational cost. Recent KAN-based methods enhance nonlinear representation learning, but most mainly use KAN as a substitute for MLP or convolutional modules, without explicitly exploiting the continuous spectral characteristics and frequency-domain structure of hyperspectral data. Moreover, increasingly complex architectures may aggravate overfitting under limited labeled samples.

To address these issues, FFKAN, a lightweight hyperspectral image classification network via frequency-domain decomposition and physically constrained Fourier KAN, is proposed in this paper. The main contributions are summarized as follows:

1) A learnable 3D wavelet frequency decomposition module (L3D-WFDM) is designed to adaptively decompose HSI features into different frequency components, enabling low-frequency structural information and high-frequency discriminative details to be captured in a learnable manner.
2) A physically constrained Fourier Kolmogorov–Arnold Network module (PC-FKAN) is developed, in which learnable Fourier basis functions are employed to model nonlinear spectral–spatial relationships, and physical constraints are introduced to encourage smooth, stable, and less redundant frequency representations under limited labeled samples.
3) An end-to-end lightweight HSI classification framework, termed FFKAN, is proposed to incorporate frequency-aware spectral–spatial modeling. Experiments on four benchmark datasets demonstrate competitive performance with fewer parameters and lower computational complexity than several state-of-the-art methods.

II. Related Work

A. Frequency Domain Analysis

Frequency domain transformation methods convert signals from the spatial or spectral domain into the frequency domain, enabling the decomposition and analysis of different frequency components. Classical techniques, such as Fourier transform, wavelet transform, and discrete cosine transform, have been widely used in signal processing and computer vision for image compression, denoising, and feature extraction [36]. In HSI classification, frequency-domain analysis is particularly useful for alleviating spectral redundancy and emphasizing discriminative spectral–spatial responses. Specifically, low-frequency components usually reflect global spectral profiles and stable spatial structures, whereas high-frequency components preserve fine-grained spectral variations, local textures, and possible noise disturbances. By distinguishing these components, models can better capture both overall material characteristics and subtle class-specific differences, thereby improving robustness while reducing unnecessary computational complexity.

Recently, frequency-domain approaches have attracted increasing attention in HSI classification. He et al. [37] proposed

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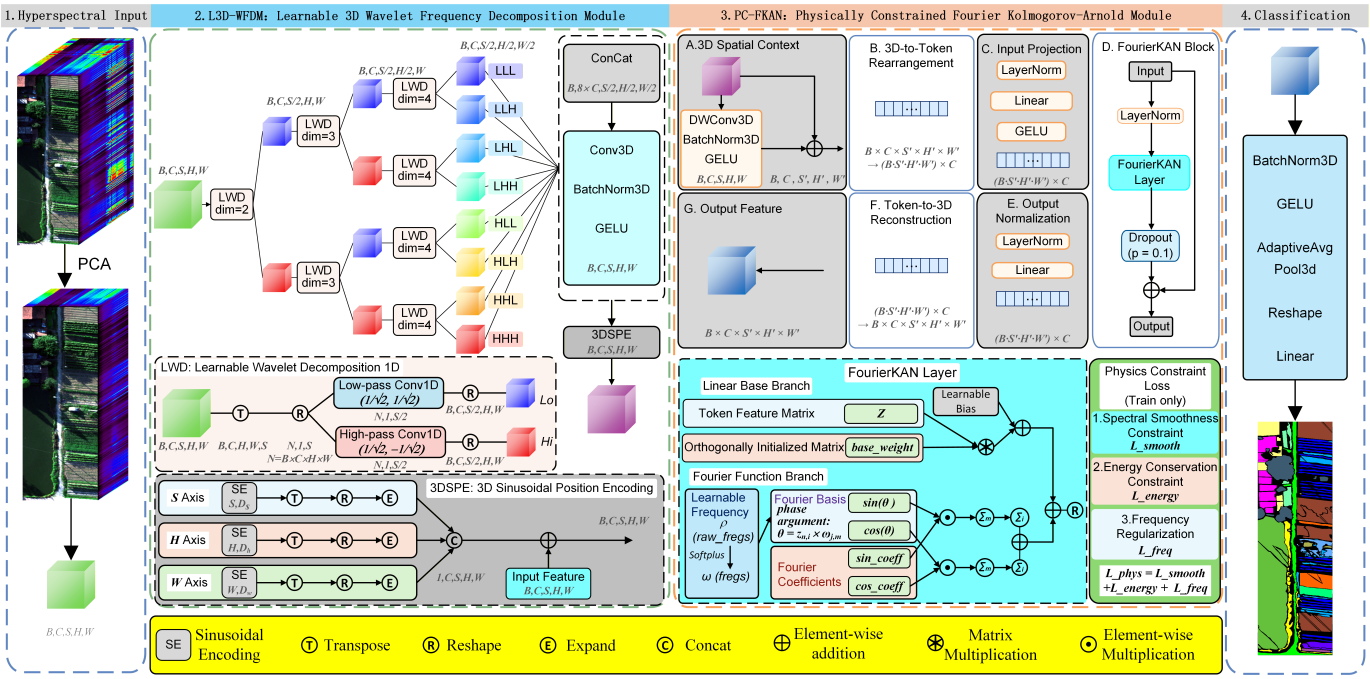


Fig. 1: Architecture of the proposed FFKAN.

a decoupled image- and frequency-domain spectral–spatial framework to enhance structural shapes and texture details. Shi et al. [38] introduced a multiscale spatial–frequency-domain cross-Transformer, which assigns distinct attention weights to FFT-based frequency bands. More recently, Zhuang et al. [39] designed a frequency-aware hierarchical Mamba network to explicitly distinguish high- and low-frequency information. These studies demonstrate the effectiveness of frequency-domain modeling for spectral–spatial representation. However, most existing methods still rely on additional frequency branches, complex attention mechanisms, or sequence models, and the integration of frequency decomposition with lightweight nonlinear representation learning remains insufficiently explored.

B. Kolmogorov–Arnold Networks

KANs are inspired by the Kolmogorov–Arnold representation theorem, which states that any multivariate continuous function on a bounded domain can be represented as a finite composition of univariate continuous functions and addition operations. For a smooth function $f : [0, 1]^n \to \mathbb{R}$, the theorem can be expressed as:

$$f(x) = f(x_1, \cdots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^{n} \phi_{q,p}(x_p) \right), \quad (1)$$

where $\phi_{q,p} : [0, 1] \to \mathbb{R}$ and $\Phi_q : \mathbb{R} \to \mathbb{R}$ are univariate functions. This theorem provides a theoretical foundation for modeling complex high-dimensional nonlinear mappings via combinations of learnable one-dimensional functions.

Inspired by this theorem, Liu et al. [31] proposed KANs as an alternative to traditional MLPs. Unlike MLPs, which apply fixed activation functions at nodes and learnable linear weights on edges, KANs assign learnable activation functions to edges and use nodes only to aggregate incoming signals. In this way, each KAN layer can be viewed as a matrix of learnable univariate functions:

$$\Phi = \{\phi_{q,p}\}, \quad p = 1, 2, \cdots, n_{\text{in}}, \quad q = 1, 2, \cdots, n_{\text{out}}. \quad (2)$$

Given the activation value $x_{l,i}$ of the i-th node in the l-th layer, the output of the j-th node in the next layer is:

$$x_{l+1,j} = \sum_{i=1}^{n_l} \phi_{l,j,i}(x_{l,i}), \quad j = 1, 2, \cdots, n_{l+1}. \quad (3)$$

Accordingly, a deep KAN can be formulated as the composition of multiple KAN layers, where $\circ$ denotes function composition:

$$\text{KAN}(x) = (\Phi_{L-1} \circ \Phi_{L-2} \circ \cdots \circ \Phi_1 \circ \Phi_0)x. \quad (4)$$

This architecture integrates linear transformations and nonlinear activations into learnable edge functions, thereby enhancing nonlinear representation capability and interpretability.

Owing to these properties, KANs show strong potential for compact and expressive nonlinear modeling. For HSI classification, they provide a promising framework for learning adaptive spectral–spatial mappings from complex material responses, subtle inter-class differences, and intra-class variations. However, existing KAN-based HSI methods mostly use KAN as a replacement for MLP layers, convolutional blocks, or classification heads, while overlooking spectral continuity and frequency characteristics. This gap motivates the proposed FFKAN, which integrates frequency decomposition with Fourier-based KAN representation for lightweight HSI classification.

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III. Methodology

A. Overall Framework

Fig. 1 illustrates the overall architecture of the proposed FFKAN. Given a hyperspectral image patch $\boldsymbol{X} \in \mathbb{R}^{B\times C\times S\times H\times W}$, where B, C, S, H, and W denote the batch size, input channel, spectral dimension, height, and width, respectively, FFKAN consists of three main components: L3D-WFDM, PC-FKAN, and a lightweight classification head.

First, L3D-WFDM transforms the input patch into frequency-aware spectral–spatial representations. It performs learnable 3D wavelet decomposition along both spectral and spatial dimensions, generating multiple frequency subbands. These subbands are then aggregated and projected into a compact feature tensor, while 3D positional encoding is introduced to preserve structural information. Second, PC-FKAN further models the decomposed features. A lightweight 3D spatial context branch first enhances local spectral–spatial dependencies. Then, the feature tensor is rearranged into token-wise representations and processed by Fourier KAN blocks to capture nonlinear frequency responses. During training, physical constraints are imposed to regularize spectral smoothness, feature energy, and frequency behavior. Finally, the classification head uses 3D batch normalization, GELU activation, adaptive average pooling, and a linear classifier to generate final classification.

B. L3D-WFDM: Learnable 3D Wavelet Frequency Decomposition Module

In HSI classification, adjacent spectral bands are highly correlated, while discriminative information often lies in both smooth spectral trends and subtle local variations. Directly modeling the original spectral–spatial cube may introduce redundancy and blur the distinction between low-frequency structures and high-frequency details. To obtain a more adaptive frequency representation, we propose L3D-WFDM, a learnable 3D wavelet frequency decomposition module.

Before L3D-WFDM, PCA is applied to reduce spectral redundancy while preserving dominant spectral information. Given an input HSI patch $\boldsymbol{X} \in \mathbb{R}^{B\times C\times S\times H\times W}$, where B, C, S, H, and W denote the batch size, input channel, spectral dimension, height, and width, respectively, L3D-WFDM decomposes $\boldsymbol{X}$ into multiple spectral–spatial frequency subbands. As shown in Fig. 1, it consists of learnable 3D wavelet frequency decomposition, lightweight subband aggregation, and 3D positional encoding.

To make wavelet decomposition adaptive to HSI data, we first design a learnable one-dimensional wavelet decomposition (LWD) operation. For a one-dimensional signal $\boldsymbol{x}$ extracted along a given dimension, LWD employs a learnable low-pass filter $\boldsymbol{h}^l$ and a learnable high-pass filter $\boldsymbol{h}^h$ to generate low-frequency and high-frequency responses. The filtering process is formulated as

$$\boldsymbol{x}^l(n) = \sum_{r=0}^{K-1} h_r^l \boldsymbol{x}(2n+r), \tag{5}$$

$$\boldsymbol{x}^h(n) = \sum_{r=0}^{K-1} h_r^h \boldsymbol{x}(2n+r), \tag{6}$$

where K denotes the filter size, h_r^l and h_r^h are learnable parameters of the low-pass and high-pass filters, respectively, and n represents the downsampled index. The stride of the filtering operation is set to 2, which simultaneously performs frequency separation and resolution reduction. In our implementation, the filters are initialized with Haar-like bases and are optimized during training:

$$\boldsymbol{h}^l = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right], \quad \boldsymbol{h}^h = \left[\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right], \tag{7}$$

This initialization provides a stable wavelet prior, while the learnable parameters allow the decomposition bases to adapt to the spectral–spatial characteristics of different HSI scenes.

Based on the LWD operation, L3D-WFDM performs separable 3D wavelet frequency decomposition along the spectral, height, and width dimensions. Instead of directly applying a fixed 3D transform, the decomposition is implemented by successively applying learnable one-dimensional wavelet filters to each axis of the input HSI patch.

Specifically, the input patch $\boldsymbol{X} \in \mathbb{R}^{B\times C\times S\times H\times W}$ is first decomposed along the spectral dimension. For each downsampled spectral index s, the spectral low-frequency and high-frequency responses are given by

$$\begin{aligned} \boldsymbol{X}_l(b,c,s,h,w) &= \sum_{r=0}^{K-1} h_r^l \boldsymbol{X}(b,c,2s+r,h,w), \\ \boldsymbol{X}_h(b,c,s,h,w) &= \sum_{r=0}^{K-1} h_r^h \boldsymbol{X}(b,c,2s+r,h,w), \end{aligned} \tag{8}$$

where $\boldsymbol{X}_l$ and $\boldsymbol{X}_h$ denote the spectral low-frequency and high-frequency components, respectively. The low-frequency component mainly captures smooth spectral reflectance trends, whereas the high-frequency component preserves subtle local spectral variations.

Then, $\boldsymbol{X}_l$ and $\boldsymbol{X}_h$ are further decomposed along the height and width dimensions, resulting in eight 3D spectral–spatial frequency subbands:

$$\begin{aligned} \mathcal{B} = \{&\boldsymbol{X}_{lll}, \boldsymbol{X}_{llh}, \boldsymbol{X}_{lhl}, \boldsymbol{X}_{lhh}, \\ &\boldsymbol{X}_{hll}, \boldsymbol{X}_{hlh}, \boldsymbol{X}_{hhl}, \boldsymbol{X}_{hhh}\}. \end{aligned} \tag{9}$$

In this notation, the three subscripts correspond to the filtering types along the spectral, height, and width dimensions, respectively, where l and h denote low-pass and high-pass filtering. From $\boldsymbol{X}_{lll}$ to $\boldsymbol{X}_{hhh}$, the eight subbands progressively describe spectral–spatial responses from low-frequency structures to high-frequency details, providing a fine-grained and discriminative frequency representation of the HSI patch.

After decomposition, the generated subbands are concatenated along the channel dimension and then aggregated by a

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lightweight $1 \times 1 \times 1$ 3D convolution followed by BN_{3D} and GELU:

$$\begin{aligned} \boldsymbol{F}_{cat} &= \mathrm{Concat}(\mathcal{B}), \\ \boldsymbol{F}_{agg} &= \mathrm{GELU}\left(\mathrm{BN}_{3D}\left(\mathrm{Conv}_{3D}^{1\times1\times1}\left(\boldsymbol{F}_{cat}\right)\right)\right). \end{aligned} \tag{10}$$

This operation mixes different frequency components and compresses the channel dimension, thereby producing a compact aggregated representation.

Although wavelet decomposition separates spectral–spatial frequency components, filtering, downsampling, and aggregation may weaken explicit positional information. Since the spectral axis retains ordered spectral-related cues, and the height and width axes describe local spatial neighborhoods, L3D-WFDM introduces 3D sinusoidal positional encoding to supplement the frequency-aware representation with coordinate cues.

Given the aggregated feature tensor $\boldsymbol{F} \in \mathbb{R}^{B\times D\times S'\times H'\times W'}$, where D denotes the output channel number and S', H', and W' denote the decomposed spectral, height, and width sizes, respectively, the channel dimension is divided into three parts:

$$\begin{aligned} D_s &= \left\lfloor \frac{D}{3} \right\rfloor, \\ D_h &= \left\lfloor \frac{D}{3} \right\rfloor, \\ D_w &= D - D_s - D_h. \end{aligned} \tag{11}$$

where D_s, D_h, and D_w denote the numbers of positional encoding channels assigned to the spectral, height, and width dimensions, respectively. This channel partition enables the model to encode positional variations along the three axes separately, instead of mixing spectral and spatial coordinates into a single ambiguous embedding.

For a one-dimensional coordinate p and encoding dimension index i, the sinusoidal positional encoding is defined as

$$\begin{aligned} PE(p, 2i) &= \sin\left(\frac{p}{10000^{2i/d}}\right), \\ PE(p, 2i+1) &= \cos\left(\frac{p}{10000^{2i/d}}\right), \end{aligned} \tag{12}$$

where d denotes the positional encoding dimension of the corresponding axis. The paired sine and cosine functions describe each coordinate from different phases at multiple frequency scales, enabling the encoding to provide both fine-grained local position cues and long-range coordinate information.

Accordingly, the positional encodings along the spectral, height, and width axes are first generated as

$$\begin{aligned} \boldsymbol{E}_s &\in \mathbb{R}^{S'\times D_s}, \\ \boldsymbol{E}_h &\in \mathbb{R}^{H'\times D_h}, \\ \boldsymbol{E}_w &\in \mathbb{R}^{W'\times D_w}. \end{aligned} \tag{13}$$

These matrices are then transposed and reshaped to align with their corresponding axes:

$$\begin{aligned} \boldsymbol{P}_s &= \mathrm{Reshape}(\boldsymbol{E}_s^T) \in \mathbb{R}^{1\times D_s\times S'\times1\times1}, \\ \boldsymbol{P}_h &= \mathrm{Reshape}(\boldsymbol{E}_h^T) \in \mathbb{R}^{1\times D_h\times1\times H'\times1}, \\ \boldsymbol{P}_w &= \mathrm{Reshape}(\boldsymbol{E}_w^T) \in \mathbb{R}^{1\times D_w\times1\times1\times W'}. \end{aligned} \tag{14}$$

The introduced singleton dimensions ensure that each positional encoding varies only along its corresponding axis. Specifically, $\boldsymbol{P}_s$ varies only along the spectral dimension and is shared across all spatial positions, $\boldsymbol{P}_h$ varies only along the height dimension and is shared across the spectral and width dimensions, and $\boldsymbol{P}_w$ varies only along the width dimension and is shared across the spectral and height dimensions.

To match the size of $\boldsymbol{F}_{agg}$, the three directional positional encodings are expanded along the remaining dimensions and concatenated along the channel dimension:

$$\begin{aligned} &\boldsymbol{P}_{3D} = [\widetilde{\boldsymbol{P}}_s, \widetilde{\boldsymbol{P}}_h, \widetilde{\boldsymbol{P}}_w] \in \mathbb{R}^{1\times D\times S'\times H'\times W'}, \\ &\widetilde{\boldsymbol{P}}_s \in \mathbb{R}^{1\times D_s\times S'\times H'\times W'}, \\ &\widetilde{\boldsymbol{P}}_h \in \mathbb{R}^{1\times D_h\times S'\times H'\times W'}, \\ &\widetilde{\boldsymbol{P}}_w \in \mathbb{R}^{1\times D_w\times S'\times H'\times W'}. \end{aligned} \tag{15}$$

Finally, the output of L3D-WFDM is obtained by adding $\boldsymbol{P}_{3D}$ to the aggregated frequency representation:

$$\boldsymbol{F}_{L3D\text{-}WFDM} = \boldsymbol{F}_{agg} + \boldsymbol{P}_{3D}. \tag{16}$$

This additive design does not change the size of the feature tensor, but explicitly supplements the frequency-aware representation with spectral–spatial coordinate information. As a result, the subsequent PC-FKAN module receives features that contain not only decomposed multi-frequency responses but also necessary structural positional cues.

C. PC-FKAN: Physically Constrained Fourier Kolmogorov–Arnold Module

After the frequency-aware decomposition of L3D-WFDM, the obtained features contain both low-frequency structural information and high-frequency discriminative details. However, these decomposed spectral–spatial responses still exhibit complex nonlinear relationships, especially when different land-cover classes share similar global spectral trends but differ in subtle local variations. To address this issue, we propose the PC-FKAN module, which introduces a Fourier-based Kolmogorov–Arnold representation to model nonlinear spectral–spatial dependencies in a lightweight and frequency-aware manner. As shown in Fig. 1, PC-FKAN consists of seven main operations: residual 3D spatial context enhancement, 3D-to-token rearrangement, input projection, Fourier KAN block, output normalization and projection, token-to-3D reconstruction, and physical constraint regularization.

PC-FKAN first introduces a lightweight residual 3D spatial context enhancement branch. Specifically, a $3\times3\times3$ depthwise 3D convolution is applied to model local neighborhoods along the spectral, height, and width dimensions independently for each channel. The convolution is followed by 3D batch normalization and GELU activation, and its output is added back to the original feature tensor through a residual connection. Given the output of L3D-WFDM, denoted as

$$F_{\mathrm{out}} \in \mathbb{R}^{B\times C\times S'\times H'\times W'}, \tag{17}$$

where B, C, S', H', and W' denote the batch size, channel dimension, decomposed spectral dimension, height, and width, respectively:

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$$F_{3DC} = \mathrm{GELU}\left(\mathrm{BN}_{3D}\left(\mathrm{DWConv}_{3D}^{3\times3\times3}(F_{\mathrm{out}})\right)\right), \\ F_c = F_{\mathrm{out}} + F_{3DC} \tag{18}$$

where F_c is the context-enhanced feature tensor. Since the depthwise convolution operates independently on each channel, this branch captures local spectral–spatial context with only a small number of additional parameters. Meanwhile, the residual connection preserves the frequency-aware representations generated by L3D-WFDM and allows the context branch to learn complementary local responses.

The context-enhanced feature tensor is then rearranged into token-wise representations. Specifically, F_c is permuted and reshaped as

$$X_t = \mathrm{Reshape}(\mathrm{Permute}(F_c)) \in \mathbb{R}^{N\times C}, \\ N = B \times S' \times H' \times W'. \tag{19}$$

In this representation, each spectral–spatial position is regarded as one token, and the channel dimension serves as its feature descriptor. This transformation allows PC-FKAN to perform token-wise nonlinear modeling while preserving the correspondence between spectral–spatial positions and their frequency-aware responses.

Before being fed into the Fourier KAN blocks, the token features are first normalized and projected into a hidden feature space through an input projection layer:

$$H_0 = \mathrm{GELU}\left(\mathrm{Linear}_{\mathrm{in}}\left(\mathrm{LN}(X_t)\right)\right), \tag{20}$$

where $\mathrm{LN}(\cdot)$ denotes layer normalization, $\mathrm{Linear}_{\mathrm{in}}(\cdot)$ maps the input channel dimension to the hidden dimension, and $H_0 \in \mathbb{R}^{N\times d_h}$. This step stabilizes the distribution of token features and introduces an initial nonlinear transformation before Fourier-domain modeling.

The projected token features are subsequently processed by a Fourier KAN block. Each block adopts a pre-normalization residual structure consisting of layer normalization, a Fourier KAN layer, dropout, and residual addition. For the l-th block, the operation is formulated as

$$H_l = H_{l-1} + \mathrm{Dropout}\left(\mathrm{FKAN}\left(\mathrm{LN}(H_{l-1})\right)\right), \\ l = 1, \ldots, L, \tag{21}$$

where the dropout probability is set to $p = 0.1$. The residual connection preserves the input token representation and allows the Fourier KAN layer to learn complementary nonlinear frequency responses.

After the Fourier KAN blocks, output normalization and projection are applied to obtain the final token representation:

$$Y_t = \mathrm{Linear}_{\mathrm{out}}\left(\mathrm{LN}(H_L)\right), \tag{22}$$

where $Y_t \in \mathbb{R}^{N\times d}$, and $\mathrm{Linear}_{\mathrm{out}}(\cdot)$ maps the hidden dimension to the output channel dimension. The token-wise features are then reconstructed into a 3D spectral–spatial tensor:

$$F_{\mathrm{pc}} = \mathrm{Reshape}^{-1}(Y_t) \in \mathbb{R}^{B\times d\times S'\times H'\times W'}. \tag{23}$$

This reconstruction restores the spectral–spatial structure of the feature tensor, enabling the subsequent classification head to perform global feature aggregation and category prediction.

In the Fourier KAN module, the Fourier KAN layer serves as the core nonlinear mapping unit. It utilizes learnable Fourier bases to model the nonlinear functions of the edge directions between the input and output channels. Given the normalized token feature $Z \in \mathbb{R}^{N\times d_{\mathrm{in}}}$, each row $z_n \in \mathbb{R}^{d_{\mathrm{in}}}$ corresponds to one spectral–spatial token. The Fourier KAN layer consists of two complementary paths: a linear base branch and a Fourier function branch. The linear base branch is formulated as:

$$Y^{\mathrm{base}} = Z \times B_{\mathrm{W}} + b, \tag{24}$$

where $B_{\mathrm{W}} \in \mathbb{R}^{d_{\mathrm{in}}\times d_{\mathrm{out}}}$ and $b \in \mathbb{R}^{d_{\mathrm{out}}}$ denote the learnable base weight and bias, respectively. This path preserves the basic linear transformation capability and provides a stable backbone during optimization.

In parallel, the Fourier function branch introduces learnable frequency bases to approximate nonlinear spectral–spatial response functions. Given the normalized token feature $Z \in \mathbb{R}^{N\times d_{\mathrm{in}}}$, where $z_{n,i} = (Z)_{n,i}$, a set of raw frequency parameters $\rho \in \mathbb{R}^{d_{\mathrm{out}}\times M}$ is first mapped to positive frequencies through the Softplus function. The Fourier basis argument is then generated by multiplying the input feature value with the learnable frequency, and the corresponding sine and cosine basis responses are computed as

$$\omega_{j,m} = \mathrm{Softplus}(\rho_{j,m}), \\ \theta_{n,i,j,m} = z_{n,i}\omega_{j,m}, \\ S_{n,i,j,m} = \sin(\theta_{n,i,j,m}), \quad C_{n,i,j,m} = \cos(\theta_{n,i,j,m}). \tag{25}$$

Here, M denotes the number of Fourier modes, while i, j, and m index the input dimension, output dimension, and Fourier mode, respectively. Through broadcasting, the basis argument tensor is formed as $\Theta \in \mathbb{R}^{N\times d_{\mathrm{in}}\times d_{\mathrm{out}}\times M}$, which contains the sine and cosine arguments for all tokens, input dimensions, output dimensions, and Fourier modes.

Meanwhile, two groups of learnable Fourier coefficients,

$$\mathrm{sin_coeff} \in \mathbb{R}^{d_{\mathrm{in}}\times d_{\mathrm{out}}\times M}, \\ \mathrm{cos_coeff} \in \mathbb{R}^{d_{\mathrm{in}}\times d_{\mathrm{out}}\times M}, \tag{26}$$

are assigned to the sine and cosine bases, respectively. The nonlinear Fourier response is obtained by aggregating all input dimensions and Fourier modes through Einstein summation:

$$Y_{n,j}^{\mathrm{fourier}} = \sum_{i=1}^{d_{\mathrm{in}}}\sum_{m=1}^{M}\Big[\mathrm{sin_coeff}_{i,j,m} \times S_{n,i,j,m} \\ + \mathrm{cos_coeff}_{i,j,m} \times C_{n,i,j,m}\Big]. \tag{27}$$

This operation corresponds to the edge-wise Fourier function mapping shown in Fig. 1, where each input–output connection is represented by a learnable Fourier function rather than a fixed scalar weight or a fixed activation function.

Finally, the output of the Fourier KAN layer is obtained by fusing the base response and the Fourier nonlinear response:

$$Y = Y^{\mathrm{base}} + Y^{\mathrm{fourier}}. \tag{28}$$

Therefore, the Fourier KAN layer combines the stability of linear projection with adaptive nonlinear frequency modeling.

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By using learnable sinusoidal bases, it can better capture smooth spectral trends and subtle local variations in hyperspectral features.

Hyperspectral representations generally exhibit local spectral continuity, while stable feature learning also requires controlled response magnitude and frequency complexity. Motivated by these characteristics, PC-FKAN introduces three physics-inspired regularization terms. Let F_{pc} denote the output feature and ω_l the learnable positive frequencies of the l-th Fourier KAN layer. For compact notation, the spectral smoothness, feature-energy, and frequency-complexity constraints are denoted by $\mathcal{L}_{\mathrm{s}}$, $\mathcal{L}_{\mathrm{e}}$, and $\mathcal{L}_{\mathrm{f}}$, respectively:

$$\begin{aligned}\mathcal{L}_{\mathrm{s}} &= \mathrm{Mean}\Big[\Big(F_{\mathrm{pc}}(s+1) - 2F_{\mathrm{pc}}(s) + F_{\mathrm{pc}}(s-1)\Big)^2\Big],\\ \mathcal{L}_{\mathrm{e}} &= \mathrm{Mean}\left(F_{\mathrm{pc}}^2\right),\\ \mathcal{L}_{\mathrm{f}} &= \frac{1}{L}\sum_{l=1}^{L}\mathrm{Mean}\left(\omega_l^2\right).\end{aligned} \tag{29}$$

where s indexes valid spectral positions and L denotes the number of Fourier KAN layers. These terms constrain spectral variations, feature magnitude, and excessive frequency complexity, respectively. Before adaptive weighting, they are normalized by their initial reference values to reduce scale differences.

Their relative contributions are learned through a bounded simplex mapping:

$$\begin{aligned}&\lambda_i = \frac{\exp\left(\tanh(a_i)/3\right)}{\sum_{k=1}^{3}\exp\left(\tanh(a_k)/3\right)}, \qquad i \in \{\mathrm{s},\mathrm{e},\mathrm{f}\},\\ &\lambda_{\mathrm{s}} + \lambda_{\mathrm{e}} + \lambda_{\mathrm{f}} = 1.\end{aligned} \tag{30}$$

where a_i is a learnable weighting parameter. The bounded mapping keeps the weights positive and normalized while allowing their relative contributions to evolve during optimization.

The overall physics-inspired constraints loss is formulated as

$$\mathcal{L}_{\mathrm{pc}} = \lambda_{\mathrm{s}}\mathcal{L}_{\mathrm{s}} + \lambda_{\mathrm{e}}\mathcal{L}_{\mathrm{e}} + \lambda_{\mathrm{f}}\mathcal{L}_{\mathrm{f}}. \tag{31}$$

Thus, PC-FKAN adaptively balances spectral smoothness, feature-energy stability, and frequency complexity for more stable frequency-aware representation learning.

Overall, PC-FKAN enhances HSI classification by modeling nonlinear responses in the frequency-aware features produced by L3D-WFDM. The learnable Fourier bases capture smooth spectral trends and subtle local variations, while the edge-wise KAN formulation improves nonlinear representation with a lightweight structure. By combining frequency-aware decomposition with physically constrained nonlinear modeling, PC-FKAN supports more robust and discriminative feature learning under limited labeled samples.

IV. Experiment and Analysis

A. Description of Hyperspectral Datasets

Four representative hyperspectral datasets were employed to comprehensively evaluate FFKAN, including Pavia University (PU), Salinas (SA), Xuzhou (XZ), and WHU-Hi-HanChuan (HC). These datasets cover urban, agricultural, mining, and peri-urban environments acquired by different airborne or unmanned aerial vehicle hyperspectral imaging systems. Specifically, PU is a 610×340 urban scene acquired by ROSIS, containing 103 retained bands, nine classes, and a spatial resolution of 1.3 m; SA is a 512×217 agricultural scene acquired by AVIRIS, containing 204 retained bands, 16 classes, and a spatial resolution of 3.7 m; XZ is a 500×260 peri-urban and mining scene acquired by an airborne HySpex system, containing 436 retained bands, nine classes, and a spatial resolution of 0.73 m; and HC is a 1217×303 high-resolution agricultural and peri-urban scene acquired by a UAV-mounted Headwall Nano-Hyperspec sensor, containing 274 bands, 16 classes, and a spatial resolution of 0.109 m.

Overall, the retained spectral dimensions range from 103 to 436 bands, the spatial resolutions vary from 3.7 m to 0.109 m per pixel, and the spectral coverage extends from approximately 400–1000 nm to 415–2508 nm. The four datasets also exhibit diverse classification challenges, including spectrally similar urban materials, subtle differences among crop categories, heterogeneous mining–urban surfaces, high-resolution spatial details, shadow interference, and severe class imbalance. These variations in sensors, spatial scales, spectral configurations, scene structures, and category distributions provide a comprehensive benchmark for evaluating the robustness and generalization capability of FFKAN. The three-band color composite images (RGB) and corresponding ground-truth (GT) maps are shown in Figs. 2–5. As reported in Table I, approximately 1% of the labeled samples from each category were randomly selected for training, while the remaining 99% were used for testing.

B. Experimental setting

All experiments were conducted on a high-performance computing platform equipped with an Intel Core i7-13700KF CPU, an NVIDIA GeForce RTX 4090 GPU with 24 GB of video memory, and 64 GB of system memory. The proposed FFKAN was implemented in Python 3.8.20 using PyTorch 2.4.1 and CUDA 12.1. For each dataset, approximately 1% of the labeled samples from each category were randomly selected for training, while the remaining 99% were used for testing, as reported in Table I. All models were trained for 100 epochs with a batch size of 64. PCA with whitening was applied to retain 10 principal components for all datasets. The Adam optimizer was adopted with initial learning rates of 0.006, 0.004, 0.003, and 0.004 for PU, SA, XZ, and HC, respectively, together with a cosine annealing learning-rate scheduler. PC-FKAN employed a single Fourier KAN layer with two Fourier modes, while the weights of the smoothness, energy, and frequency regularization terms were treated as learnable parameters and optimized during training. To reduce the influence of random sample selection and parameter initialization, the main comparative experiments were independently repeated five times using different random seeds, and the quantitative results are reported as mean $\pm$ standard deviation over the five runs. Unless otherwise specified, the comparative methods were implemented using the parameter configurations recommended in their original publications.

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To comprehensively evaluate the classification performance and computational efficiency of FFKAN, seven representative methods were selected for comparison, including the Transformer-based SSFTT [21] and MASSFormer [40], the Mamba-based S^{2}Mamba [41], and four lightweight methods, namely LSGA [42], SYNet [43], IGroupSS-Mamba [25], and SliMamba [44]. These methods cover Transformer, state-space, convolutional, attention-based, and lightweight architectures, enabling a comprehensive assessment of FFKAN in terms of classification accuracy, parameter efficiency, and adaptability under limited-training-sample conditions.

Classification performance was comprehensively assessed using three standard metrics: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ). Specifically, OA measures the ratio of correctly classified samples to the total number of test samples, AA is computed as the mean of class-wise accuracies to reflect balanced performance across categories, and κ quantifies the agreement between predicted and ground-truth labels while accounting for chance agreement. Their formulations are defined as:

$$\begin{aligned} \mathrm{OA} &= \frac{\sum_{i=1}^{C} n_{ii}}{N} \\ \mathrm{AA} &= \frac{1}{C}\sum_{i=1}^{C}\frac{n_{ii}}{n_i} \\ \kappa &= \frac{P_o - P_e}{1 - P_e}, \quad P_o = \mathrm{OA}, \quad P_e = \frac{\sum_{i=1}^{C} n_i n_i'}{N^2} \end{aligned} \tag{32}$$

where C denotes the number of classes, N is the total number of test samples, n_{ii} represents the number of correctly classified samples of class i, and n_i and n_i' denote the numbers of ground-truth and predicted samples in class i, respectively.

C. Analysis of Experimental Results

The quantitative comparisons between FFKAN and seven representative state-of-the-art methods on the Pavia University, Salinas, Xuzhou, and WHU-Hi-HanChuan datasets are presented in Table II, Table III, Table IV, and Table V, respectively. In these tables, the best result for each class and evaluation metric is highlighted in bold. Furthermore, to provide an intuitive visual comparison of the spatial classification performance, the classification maps shown in Figs. 2–5 are generated from a randomly selected run among the five independent experiments.

1) Pavia University Dataset: Table II reports the quantitative comparison on the Pavia University dataset. FFKAN achieves the best overall performance with an OA of 98.54%, an AA of 96.91%, and a κ coefficient of 98.06%. Compared with S^{2}Mamba, which obtains the second-best OA and κ, FFKAN improves these two metrics by 0.51 and 0.67 percentage points, respectively. It also surpasses MASSFormer by 0.75 percentage points in AA. Moreover, FFKAN exhibits superior stability, with OA and κ standard deviations of only 0.03 across five independent runs, demonstrating reliable classification performance.

At the class level, FFKAN achieves the highest accuracy on Gravel, reaching 96.33%, while obtaining competitive accuracies of 93.83% and 95.69% on Trees and Shadows, respectively. Gravel is particularly challenging due to its spectral similarity with other urban materials. While SSFTT, IGroupSS-Mamba, LSGA, and MASSFormer obtain 85.41%, 86.39%, 90.67%, and 90.70%, FFKAN improves this category by 5.63 percentage points over the second-best method. It also obtains competitive performance on Self-Bricks with 97.64%, indicating effective discrimination of similar urban surfaces. However, FFKAN remains less effective for Asphalt and Bitumen, suggesting that some highly correlated man-made

TABLE I: Training and test sample counts for each category in the Pavia University, Salinas, Xuzhou, and WHU-Hi-HanChuan datasets.

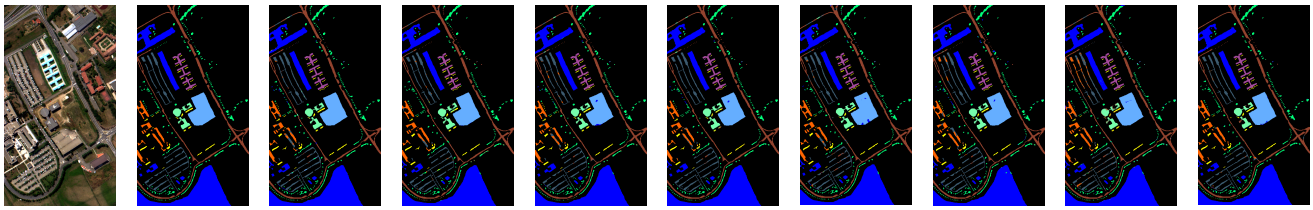
		Pavia University				**Salinas**				**Xuzhou**				**WHU-Hi-HanChuan**			
NO.	Color	Class	Train(1%)	Test(99%)	Total	Class	Train(1%)	Test(99%)	Total	Class	Train(1%)	Test(99%)	Total	Class	Train(1%)	Test(99%)	Total
1		Asphalt	66	6565	6631	Brocoli-1	20	1989	2009	BareLand-1	263	26133	26396	Strawberry	447	44288	44735
2		Meadows	186	18463	18649	Brocoli-2	37	3689	3726	Lakes	40	3987	4027	Cowpea	227	22526	22753
3		Gravel	21	2078	2099	Fallow	20	1956	1976	Coals	28	2755	2783	Soybean	103	10184	10287
4		Trees	31	3033	3064	Fallow-Rough	14	1380	1394	Cement	52	5162	5214	Sorghum	54	5299	5353
5		Painted metal	13	1332	1345	Fallow-Smooth	27	2651	2678	Crops-1	132	13052	13184	Water Spinach	12	1188	1200
6		Bare Soil	50	4979	5029	Stubble	39	3920	3959	Trees	24	2412	2436	Watermelon	45	4488	4533
7		Bitumen	13	1317	1330	Celery	36	3543	3579	Bareland-2	70	6920	6990	Greens	59	5844	5903
8		Self-Bricks	37	3645	3682	Grapes-Un.	113	11158	11271	Crops-2	48	4729	4777	Trees	180	17798	17978
9		Shadows	10	937	947	Soil-Vin.-Dev	62	6141	6203	Red tiles	31	3039	3070	Grass	95	9374	9469
10		–	–	–	–	Corn-Green	33	3245	3278	–	–	–	–	Red Roof	105	10411	10516
11		–	–	–	–	Lettuce-4wk	11	1057	1068	–	–	–	–	Gray Roof	169	16742	16911
12		–	–	–	–	Lettuce-5wk	19	1908	1927	–	–	–	–	Plastic	37	3642	3679
13		–	–	–	–	Lettuce-6wk	9	907	916	–	–	–	–	Bare soil	91	9025	9116
14		–	–	–	–	Lettuce-7wk	11	1059	1070	–	–	–	–	Road	186	18374	18560
15		–	–	–	–	Vinyard-Un.	72	7196	7268	–	–	–	–	Bright Object	11	1125	1136
16		–	–	–	–	Vinyard-Vert.	18	1789	1807	–	–	–	–	Water	754	74647	75401
		Total	427	42349	42776	Total	541	53588	54129	Total	688	68189	68877	Total	2575	254955	257530

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TABLE II: Accuracy of each model for each class (%), OA (%), AA (%), and κ (%) on the Pavia University dataset

NO.	SSFTT	LSGA	MASSFormer	S^{2}Mamba	SYNet	IGroupSS-Mamba	SliMamba	FFKAN
1	96.63±0.83	99.23±0.62	98.27±0.46	**99.62±0.40**	96.42±1.37	98.60±0.32	95.43±0.96	99.12±0.48
2	99.92±0.05	99.83±0.07	**99.96±0.04**	99.93±0.09	99.93±0.05	99.81±0.06	99.69±0.07	99.91±0.04
3	85.41±2.89	90.67±2.68	90.70±3.21	88.69±6.77	87.93±5.17	86.39±3.78	84.13±4.36	**96.33±0.71**
4	**94.11±1.23**	88.95±1.05	91.92±0.62	92.77±0.64	91.64±1.49	90.21±1.19	89.84±3.58	93.83±0.74
5	99.64±0.13	97.92±0.98	98.84±0.56	**99.96±0.08**	97.71±1.24	99.50±0.20	99.56±0.35	98.96±0.32
6	99.28±0.69	**99.97±0.07**	98.23±0.21	99.09±1.25	99.35±0.54	99.42±0.56	95.73±2.76	99.50±0.24
7	99.80±0.19	**99.83±0.17**	99.72±0.41	98.06±5.58	99.70±0.46	99.64±0.50	90.22±2.98	91.23±1.96
8	92.17±1.03	**98.08±0.51**	93.26±1.32	96.00±0.98	92.90±2.14	92.26±1.22	77.29±5.59	97.64±0.75
9	93.93±2.19	84.85±2.71	94.50±3.31	86.81±10.39	91.62±3.61	**95.74±2.69**	94.10±4.27	95.69±0.92
OA (%)	97.39±0.29	97.98±0.19	97.72±0.08	98.03±0.45	97.27±0.54	97.48±0.26	94.75±0.80	**98.54±0.03**
AA (%)	95.65±0.49	95.48±0.55	96.16±0.13	95.66±1.50	95.24±0.89	95.73±0.65	91.78±1.29	**96.91±0.23**
κ (%)	96.54±0.38	97.32±0.25	96.98±0.11	97.39±0.60	96.38±0.72	96.65±0.34	93.01±1.07	**98.06±0.03**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 2: A comparison of classification outcomes produced by various models on the Pavia University dataset. (a) False-color image, (b) Ground Truth map, (c) SSFTT (OA=97.37%), (d) LSGA (OA=97.98%), (e) MASSFormer (OA=97.71%), (f) S^{2}Mamba (OA=98.09%), (g) SYNet (OA=97.28%), (h) IGroupSS-Mamba (OA=97.43%), (i) SliMamba (OA=95.13%), and (j) FFKAN (OA=98.57%).

TABLE III: Accuracy of each model for each class (%), OA (%), AA (%), and κ (%) on the Salinas dataset

NO.	SSFTT	LSGA	MASSFormer	S^{2}Mamba	SYNet	IGroupSS-Mamba	SliMamba	FFKAN
1	99.99±0.02	99.90±0.13	99.97±0.06	99.97±0.04	**100.00±0.00**	**100.00±0.01**	97.54±2.23	99.49±0.25
2	99.98±0.04	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.94±0.05	99.98±0.03
3	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.86±0.25	**100.00±0.00**	97.84±2.31	99.92±0.10
4	98.58±3.24	99.62±0.31	**99.86±0.15**	99.75±0.14	97.61±0.72	99.68±0.13	97.68±1.03	98.35±1.41
5	99.29±0.32	98.21±0.55	99.62±0.19	99.47±0.51	99.35±0.56	**99.84±0.13**	97.35±1.30	99.47±0.30
6	99.88±0.11	**100.00±0.00**	**100.00±0.01**	**100.00±0.00**	99.93±0.12	99.98±0.01	99.83±0.16	99.98±0.02
7	99.89±0.28	99.95±0.08	99.96±0.07	**99.97±0.03**	99.96±0.05	99.95±0.02	99.05±0.66	99.02±0.13
8	99.15±0.44	99.28±0.22	99.12±0.20	99.12±0.25	99.43±0.35	98.51±0.27	90.44±2.02	**99.88±0.07**
9	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.01**	**100.00±0.00**	99.64±0.62	**100.00±0.00**
10	99.52±0.34	99.67±0.15	99.62±0.30	99.58±0.19	99.65±0.33	98.82±0.31	95.34±1.38	**99.78±0.12**
11	**99.96±0.10**	99.73±0.11	99.69±0.40	**99.96±0.08**	99.40±0.67	99.23±0.83	91.20±4.29	99.56±0.17
12	99.30±1.47	99.99±0.05	99.78±0.32	**100.00±0.00**	99.90±0.16	99.99±0.03	99.50±0.39	99.87±0.17
13	94.76±1.58	90.10±6.76	93.72±3.56	93.40±2.11	93.00±3.26	94.94±1.97	94.62±3.88	**95.00±1.62**
14	99.23±0.18	96.52±2.50	98.90±0.33	98.43±1.01	**99.29±0.52**	98.91±0.33	94.27±1.99	98.25±0.52
15	97.67±0.63	97.88±0.62	97.25±1.04	97.23±1.10	98.96±0.35	97.25±0.60	86.32±3.91	**99.44±0.27**
16	99.75±0.25	99.95±0.08	99.94±0.09	99.93±0.11	99.81±0.16	99.50±0.11	96.73±2.33	**100.00±0.00**
OA (%)	99.25±0.02	99.20±0.14	99.25±0.17	99.24±0.18	99.46±0.12	99.09±0.10	94.90±1.02	**99.60±0.05**
AA (%)	99.18±0.19	98.80±0.43	99.21±0.22	99.18±0.18	99.14±0.21	99.16±0.15	96.08±0.86	**99.25±0.15**
κ (%)	99.17±0.02	99.10±0.15	99.17±0.19	99.15±0.21	99.40±0.13	98.99±0.11	94.32±1.14	**99.55±0.06**

materials are still difficult to separate.

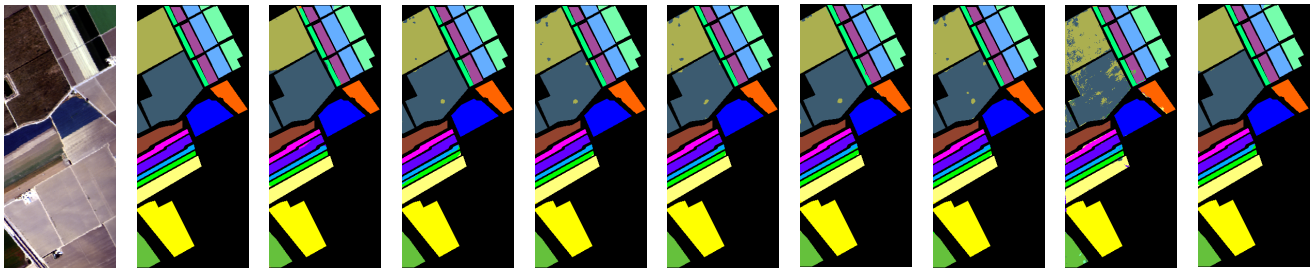
The visual results in Fig. 2 further verify these observations. Most methods recover large homogeneous regions, but errors appear around narrow structures and material boundaries. SliMamba produces more fragmented predictions, while SSFTT and IGroupSS-Mamba show confusion in Gravel and Self-Bricks regions. In comparison, FFKAN preserves clearer boundaries and reduces scattered errors in complex urban areas. The selected FFKAN result achieves an OA of 98.57%, close to the five-run average, confirming its effectiveness for urban scenes with spectral similarity and complex spatial structures.

2) Salinas Dataset: Table III presents the quantitative comparison on the Salinas dataset. FFKAN obtains the highest OA, AA, and κ values of 99.60%, 99.25%, and 99.55%, respectively. Compared with SSFTT and MASSFormer, both of which achieve an OA of 99.25%, FFKAN improves the overall accuracy by 0.35 percentage points. It also surpasses MASSFormer, which achieves an AA of 99.21%, by 0.04 percentage points and improves the κ coefficient by 0.38 percentage points over both SSFTT and MASSFormer.

Regarding individual categories, FFKAN shows clear advantages on spectrally similar agricultural classes. For Grapes_untrained and Vinyard_untrained, it achieves the

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 3: A comparison of classification outcomes produced by various models on the Salinas dataset. (a) False-color image, (b) Ground Truth map, (c) SSFTT (OA=99.21%), (d) LSGA (OA=99.21%), (e) MASSFormer (OA=99.26%), (f) S^{2}Mamba (OA=99.27%), (g) SYNet (OA=99.44%), (h) IGroupSS-Mamba (OA=99.11%), (i) SliMamba (OA=94.83%), and (j) FFKAN (OA=99.61%).

TABLE IV: Accuracy of each model for each class (%), OA (%), AA (%), and κ (%) on the Xuzhou dataset

NO.	SSFTT	LSGA	MASSFormer	S^{2}Mamba	SYNet	IGroupSS-Mamba	SliMamba	FFKAN
1	97.85±0.24	98.21±0.14	98.68±0.25	97.94±0.15	97.97±0.25	**99.14±0.08**	96.80±0.50	98.87±0.14
2	99.28±0.34	**99.82±0.07**	99.16±0.24	99.69±0.12	99.56±0.39	99.42±0.15	98.33±1.35	99.41±0.09
3	97.72±0.75	98.37±0.51	98.33±0.44	94.51±1.13	98.10±0.95	**99.01±0.15**	95.06±3.20	98.74±0.26
4	98.40±0.31	**99.12±0.28**	98.22±0.62	98.22±0.39	98.58±0.83	96.13±1.01	96.22±1.26	98.35±0.36
5	99.49±0.10	99.35±0.11	99.39±0.20	99.36±0.25	99.14±0.27	98.94±0.21	98.68±0.39	**99.51±0.08**
6	99.20±0.43	98.88±0.32	98.88±0.40	95.34±0.80	99.58±0.18	99.46±0.22	87.25±3.88	**99.68±0.19**
7	**99.20±0.23**	98.07±0.36	99.02±0.15	97.72±0.25	98.87±0.40	98.96±0.15	94.45±3.06	99.04±0.30
8	99.28±0.41	**100.00±0.01**	98.93±0.38	99.36±0.63	99.93±0.13	99.55±0.11	97.36±0.91	99.92±0.08
9	97.11±0.26	96.10±0.80	**97.41±0.61**	97.36±0.65	96.41±0.61	96.62±0.32	93.89±1.19	95.59±0.86
OA (%)	98.54±0.12	98.64±0.06	98.80±0.13	98.15±0.15	98.55±0.12	98.80±0.06	96.47±0.57	**98.95±0.03**
AA (%)	98.62±0.12	98.66±0.09	98.67±0.15	97.72±0.24	98.68±0.17	98.58±0.10	95.34±0.69	**98.79±0.10**
κ (%)	98.15±0.16	98.27±0.08	98.48±0.17	97.66±0.19	98.16±0.15	98.47±0.08	95.52±0.72	**98.67±0.03**

best accuracies of 99.88% and 99.44%, respectively. In particular, the accuracy of Vinyard_untrained is 0.48 percentage points higher than SYNet, while SliMamba obtains only 86.32%, revealing its difficulty in distinguishing vineyard-related categories. FFKAN also performs best on Corn_senesced_green_weeds, Lettuce_romaine_6wk, and Vinyard_vertical_trellis, demonstrating stronger discrimination among crops with similar spectral responses. However, several categories, including Fallow_rough_plow, Celery, and Lettuce_romaine_7wk, remain relatively challenging.

The visual results in Fig. 3 further confirm these findings. Most methods recover the major agricultural fields, but errors mainly occur near field boundaries and adjacent crop regions. SliMamba produces severe fragmented predictions in large fields, while MASSFormer and S^{2}Mamba still show local category intrusion. In contrast, FFKAN preserves clearer field structures and reduces scattered errors, especially in vineyard regions and elongated crop strips. The selected FFKAN result achieves an OA of 99.61%, consistent with the five-run average, demonstrating reliable classification for large-scale agricultural scenes.

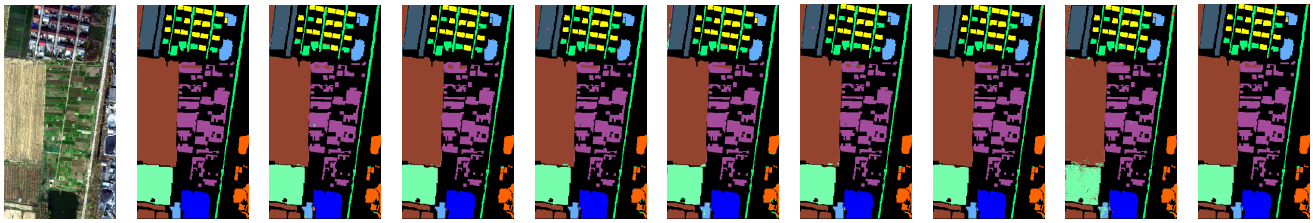
3) Xuzhou Dataset: Table IV summarizes the results on the Xuzhou dataset, which contains heterogeneous mining, agricultural, and urban land-cover types. FFKAN achieves the highest OA, AA, and κ values of 98.95%, 98.79%, and 98.67%, respectively. Although the performance gaps among the top-ranked methods are relatively small, FFKAN consistently outperforms strong competitors such as MASSFormer and IGroupSS-Mamba, both of which obtain an OA of 98.80%. In addition, its lower standard deviations of OA and κ indicate more stable behavior across repeated experiments. These results demonstrate that FFKAN can effectively adapt to complex scenes containing diverse land-cover structures.

The class-wise results further reveal its balanced recognition capability. FFKAN performs particularly well on Crops-1, Trees, and Crops-2, obtaining accuracies of 99.51%, 99.68%, and 99.92%, respectively. For Coals, which represents a typical mining material with complex spectral characteristics, FFKAN reaches 98.74%, clearly outperforming S^{2}Mamba and SliMamba. Meanwhile, several categories remain challenging for all methods. Red tiles show relatively lower accuracies, which may be associated with similar spectral responses between artificial materials and surrounding surfaces. In addition, categories with limited training samples are more sensitive to spatial variations, making accurate recognition difficult. Despite the relatively lower accuracy on Red tiles, FFKAN maintains strong performance across most other categories, indicating its adaptability to heterogeneous spectral–spatial distributions.

The visual comparisons in Fig. 4 agree with the quantitative results. Most methods recover the major regions, but errors frequently appear around fragmented objects, narrow structures, and transitions between mining and urban areas. SliMamba produces more scattered predictions, especially in small patches and boundary regions, while other methods still exhibit local category intrusion. In contrast, FFKAN

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 4: A comparison of classification outcomes produced by various models on the Xuzhou dataset. (a) False-color image, (b) Ground Truth map, (c) SSFTT (OA=98.52%), (d) LSGA (OA=98.65%), (e) MASSFormer (OA=98.80%), (f) S^2Mamba (OA=98.15%), (g) SYNet (OA=98.52%), (h) IGroupSS-Mamba (OA=98.80%), (i) SliMamba (OA=96.51%), and (j) FFKAN (OA=98.96%).

TABLE V: Accuracy of each model for each class (%), OA (%), AA (%), and κ (%) on the WHU-Hi-HanChuan dataset

NO.	SSFTT	LSGA	MASSFormer	S^2Mamba	SYNet	IGroupSS-Mamba	SliMamba	FFKAN
1	99.61±0.11	99.52±0.12	99.61±0.07	99.44±0.07	99.44±0.17	99.18±0.13	97.87±0.44	**99.93±0.05**
2	96.48±0.79	96.95±0.38	95.45±0.48	97.37±0.38	95.08±0.69	95.63±0.72	90.70±1.63	**98.66±0.50**
3	99.35±0.24	99.42±0.24	97.51±0.67	**99.63±0.15**	99.05±0.70	96.77±0.39	91.50±2.11	98.97±0.41
4	99.25±0.24	99.07±0.34	**99.30±0.19**	98.87±0.30	98.77±0.62	98.50±0.26	97.44±0.88	99.09±0.30
5	97.34±1.13	97.94±0.98	90.95±3.30	96.21±1.12	95.83±2.55	86.92±2.04	79.78±4.07	**98.87±0.72**
6	**93.44±0.77**	93.10±0.87	84.90±1.43	88.54±0.99	92.75±1.54	83.98±1.81	68.43±3.07	90.70±1.30
7	94.36±1.01	94.47±0.59	92.78±0.78	92.69±0.38	95.02±1.45	92.39±0.71	89.03±1.75	**95.91±0.73**
8	97.06±0.37	**97.41±0.28**	96.74±0.55	96.81±0.62	96.76±0.80	96.29±0.39	90.36±2.27	96.66±0.48
9	96.76±0.45	96.27±0.49	98.01±0.55	**98.62±0.23**	95.45±1.01	96.80±0.74	90.90±1.69	97.29±0.89
10	98.86±0.34	98.51±0.18	99.30±0.23	**99.69±0.07**	98.64±0.50	99.40±0.16	97.52±0.57	97.94±0.53
11	98.22±0.28	98.15±0.15	98.68±0.26	98.85±0.13	98.09±0.37	98.80±0.27	96.59±0.83	**99.09±0.29**
12	98.44±0.61	**99.03±0.49**	91.58±2.50	96.71±0.74	97.25±1.42	92.65±1.07	73.94±3.61	98.95±0.56
13	88.09±0.69	89.70±0.42	86.37±1.26	88.34±0.65	84.98±1.40	84.79±1.33	71.99±2.73	**92.30±0.76**
14	98.02±0.39	98.08±0.28	96.84±0.23	97.58±0.54	97.99±0.54	96.19±0.27	91.88±1.73	**98.14±0.64**
15	77.69±5.52	75.37±4.35	79.62±4.72	81.58±1.97	**83.03±3.96**	77.29±1.58	72.69±5.25	71.79±1.91
16	**99.95±0.03**	99.92±0.02	99.90±0.03	99.91±0.02	99.89±0.04	99.89±0.02	99.56±0.15	99.91±0.03
OA (%)	98.13±0.11	98.20±0.06	97.58±0.13	98.14±0.09	97.75±0.19	97.26±0.17	93.98±0.22	**98.50±0.06**
AA (%)	95.81±0.45	95.81±0.25	94.22±0.46	95.68±0.20	95.50±0.52	93.46±0.35	87.51±0.61	**95.89±0.13**
κ (%)	97.81±0.13	97.89±0.07	97.16±0.15	97.83±0.10	97.36±0.23	96.79±0.20	92.95±0.26	**98.24±0.07**

preserves clearer spatial structures and reduces isolated errors in heterogeneous regions. The selected result achieves an OA of 98.96%, close to the average performance, confirming its effectiveness in complex mixed scenes.

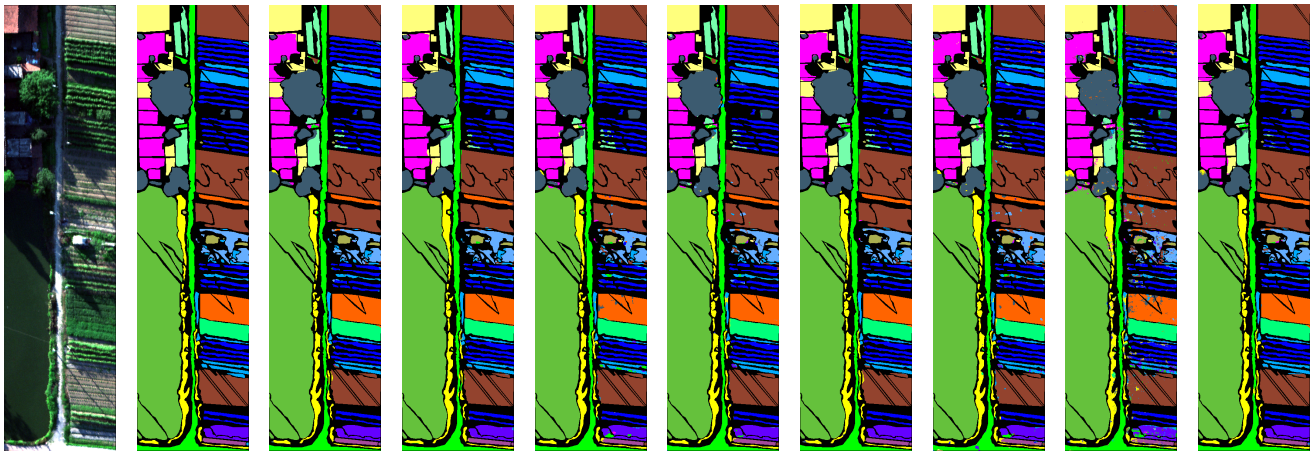
4) WHU-Hi-HanChuan Dataset: Table V presents the comparison results on the WHU-Hi-HanChuan dataset, which contains complex agricultural structures with high spatial resolution and notable class imbalance. FFKAN achieves the best overall performance, with OA, AA, and κ values of 98.50%, 95.89%, and 98.24%, respectively. Several methods also obtain competitive results on this dataset. For example, LSGA reaches an OA of 98.20% and a κ coefficient of 97.89%, while SSFTT and S^2Mamba achieve OAs of 98.13% and 98.14%, respectively. Compared with these strong competitors, FFKAN maintains a consistent advantage in all overall metrics. Moreover, its OA and κ standard deviations are only 0.06 and 0.07, indicating stable classification performance across repeated experiments. These results suggest that FFKAN can effectively extract discriminative spectral–spatial representations under complex high-resolution agricultural scenes.

The category-level results further demonstrate the robustness of FFKAN on difficult classes. It achieves the best performance on Cowpea, Water Spinach, Greens, Bare soil, and several other categories. For example, FFKAN reaches 98.66% for Cowpea and 98.87% for Water Spinach, while several competing methods show larger performance degradation. A notable improvement is observed for Bare soil, where FFKAN obtains 92.30%, exceeding most compared methods and contributing to its highest AA. Nevertheless, some categories remain challenging for all approaches. Bright Object achieves only 71.79% with FFKAN, and most competing methods also suffer from low accuracy, which may be related to its limited training samples and scattered spatial distribution. Similarly, Watermelon remains difficult because of spectral similarity with neighboring vegetation categories. These results suggest that small-scale objects and spectrally related classes remain challenging even with advanced spectral–spatial modeling.

As shown in Fig. 5, most methods successfully classify large agricultural regions, but errors mainly occur around fragmented targets, narrow boundaries, and small objects. SliMamba produces severe scattered predictions, whereas other methods still exhibit local category intrusion. In comparison, FFKAN preserves more continuous structures and reduces isolated errors, particularly in complex field regions. The selected FFKAN result achieves an OA of 98.53%, confirming its robustness in high-resolution agricultural scenes.

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 5: A comparison of classification outcomes produced by various models on the WHU-Hi-HanChuan dataset. (a) False-color image, (b) Ground Truth map, (c) SSFTT (OA=98.14%), (d) LSGA (OA=98.15%), (e) MASSFormer (OA=97.64%), (f) S^{2}Mamba (OA=98.12%), (g) SYNet (OA=97.69%), (h) IGroupSS-Mamba (OA=97.29%), (i) SliMamba (OA=93.99%), and (j) FFKAN (OA=98.53%).

D. Parameter Sensitivity Analysis

For the parameter sensitivity experiments in this section, stratified five-fold cross-validation was conducted within the 1% labeled training samples. In each fold, approximately 80% of the samples were used for training and the remaining 20% for validation. Five fold-wise experiments were performed for each parameter setting, and the resulting OA, AA, and κ were averaged for evaluation.

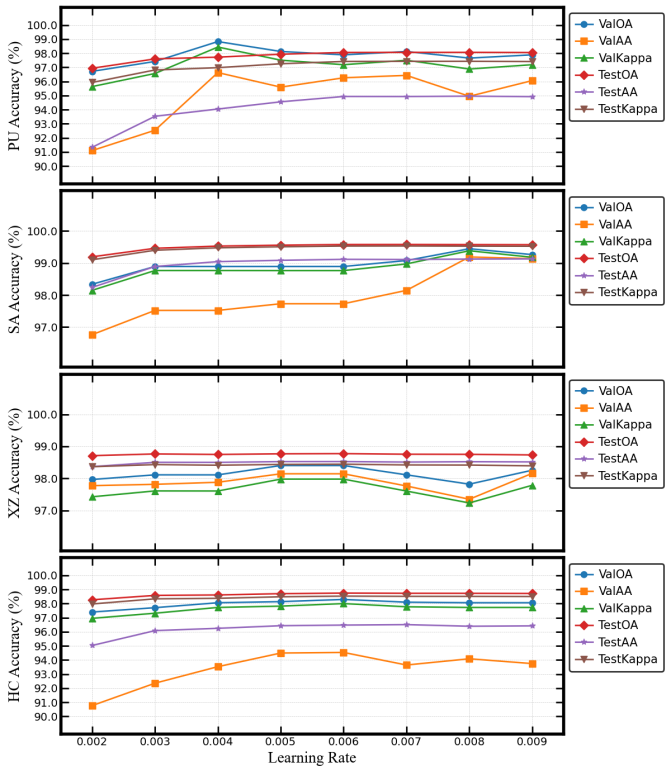


Fig. 6: Analysis of learning rate effects.

1) Learning Rate Sensitivity on OA, AA, and κ: The learning rate directly affects parameter updates and model optimization. As shown in Fig. 6, relatively small learning rates generally lead to lower OA, AA, and κ, while increasing the learning rate improves performance before the curves fluctuate or gradually saturate. The validation results are more sensitive to learning-rate changes, whereas the test results remain comparatively stable over moderate ranges. For PU, all validation metrics peak at 0.004, while the best test OA appears at 0.008, although the test performance is nearly saturated from 0.006 to 0.009. For SA, the validation optimum is 0.008, whereas the highest test OA occurs at 0.007. On XZ, validation OA and κ peak at 0.005/0.006, while the test metrics favor 0.006. HC shows the highest consistency, with both validation and test OA peaking at 0.006.

2) Patch Size Sensitivity on OA, AA, and κ: The patch size determines the spatial context available for spectral–spatial representation learning. A small patch may provide insufficient neighborhood information, whereas an excessively large patch may introduce redundant background responses. As shown in Fig. 7, the validation results generally exhibit larger variations with patch size, while the test results are relatively smoother. For PU, the validation OA, AA, and κ simultaneously peak at 17×17, reaching 98.83%, 96.63%, and 98.44%, respectively, whereas the best test OA is obtained at 13×13. For SA, all validation metrics favor 27×27, while the highest test OA occurs at 25×25; however, their test OA differs by only 0.0015 percentage points, indicating a broad performance plateau. On XZ, the best validation OA is achieved at 15×15, while the test optimum appears at 13×13, with only a 0.03 percentage-point difference in OA. HC benefits from broader spatial contexts: the validation OA peaks at 37×37, whereas the test OA reaches its maximum at 43×43. Nevertheless, the improvement beyond 37×37 is limited and the validation performance has already saturated.

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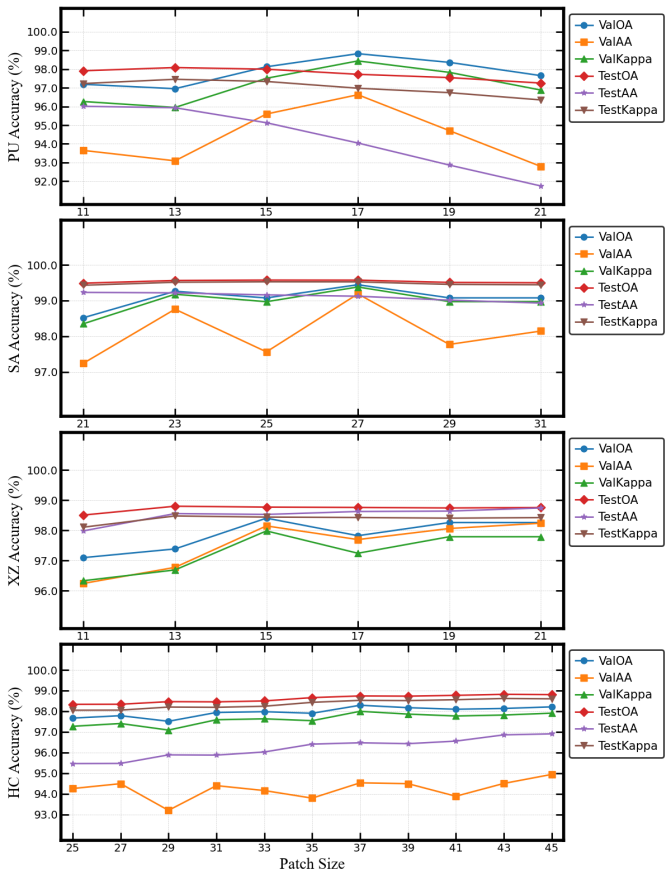


Fig. 7: Analysis of patch size effects.

3) Sensitivity analysis of PC-FKAN mode number and network depth: The Fourier mode number M determines the frequency representation capacity of each Fourier KAN layer, while the network depth L controls the number of stacked Fourier KAN blocks. Their settings directly affect the representation capability and computational efficiency of PC-FKAN. As reported in Table VI, when $L = 1$, increasing M from 1 to 2 improves OA, AA, and κ from 97.85%, 95.44%, and 97.15% to 98.11%, 96.03%, and 97.49%, respectively. However, further increasing M to 4 or 8 provides no additional performance gain, while the parameter count and runtime increase substantially. A similar trend is observed for network depth. With $M = 2$, increasing L from 1 to 3 enlarges the model from 9.13 K to 19.69 K parameters and noticeably increases the training and testing time, whereas the classification performance remains nearly unchanged or slightly decreases.

TABLE VI: Sensitivity analysis of M and L on the Pavia University dataset.

M	L	OA (%)	AA (%)	κ (%)	Param(K)	Train(s)	Test(s)
1	1	97.85 ± 0.26	95.44 ± 0.63	97.15 ± 0.34	7.05	5.73	1.79
2	**1**	$\mathbf{98.11 \pm 0.17}$	$\mathbf{96.03 \pm 0.54}$	$\mathbf{97.49 \pm 0.22}$	9.13	6.52	2.41
4	1	97.95 ± 0.20	95.76 ± 0.44	97.28 ± 0.26	13.29	9.76	3.08
8	1	98.09 ± 0.16	95.96 ± 0.52	97.47 ± 0.21	21.61	17.08	5.29
2	2	98.08 ± 0.17	95.83 ± 0.62	97.45 ± 0.23	14.41	9.84	4.02
2	3	98.03 ± 0.20	95.75 ± 0.57	97.39 ± 0.26	19.69	12.81	5.58

Overall, $M = 2$ and $L = 1$ provide a favorable balance between representation capability, classification accuracy, and computational efficiency, and are therefore adopted in FFKAN.

V. Discussion

A. Ablation experiments of network components

To systematically assess the contribution of each component in FFKAN, a stepwise ablation study was conducted on the Pavia University, Salinas, Xuzhou, and WHU-Hi-HanChuan datasets. The proposed network consists of two main modules: L3D-WFDM and PC-FKAN. Specifically, L3D-WFDM contains 3D frequency decomposition (FD) and 3D sinusoidal positional encoding (3DSPE), while PC-FKAN comprises the Fourier KAN backbone (FKAN) and the physics-inspired constraints (PC). Eight configurations were constructed by selectively introducing or removing these components. All experiments were independently repeated five times under identical settings, and the results are reported as mean $\pm$ standard deviation in Table VII.

The components of L3D-WFDM are evaluated in cases 1 and 2, whereas cases 4 and 5 examine FKAN with and without PC. Components from the two modules are further combined in cases 3, 6, and 7, while case 8 represents the complete FFKAN. The comparison between cases 1 and 2 shows that adding 3DSPE consistently improves OA, AA, and κ on all four datasets. This confirms that explicit positional information helps preserve the spectral–spatial organization of the decomposed frequency features. A much larger improvement is obtained after introducing FKAN. Compared with the FD-based configurations, cases 3 and 7 achieve substantially higher classification accuracies, demonstrating that FKAN effectively models the nonlinear dependencies among frequency-aware spectral–spatial features. In addition, case 7 consistently outperforms the standalone FKAN configuration in case 4, indicating that the complete L3D-WFDM provides complementary low- and high-frequency information for subsequent nonlinear representation learning. The effectiveness of PC is demonstrated by the comparisons between cases 4 and 5, cases 3 and 6, and cases 7 and 8. Adding PC generally improves OA, AA, and κ while maintaining small standard deviations, suggesting that the physically inspired regularization promotes more stable and less redundant frequency representations during optimization.

Overall, case 8 achieves the best results on all four datasets. These results verify that FD, 3DSPE, FKAN, and PC contribute from different perspectives and work complementarily to produce more discriminative, stable, and robust spectral–spatial representations.

B. Computational Complexity Analysis

The spatial patch size fundamentally determines the computational complexity of the classification pipeline by dictating the spatial boundary of each input cube and the tensor dimensions of intermediate feature maps. While expanding the patch size generally increases theoretical FLOPs, memory footprint, and actual execution time, the parameter count remains invariant. Therefore, all comparative methods were

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TABLE VII: Ablation study of the proposed components across four benchmark HSI datasets Mean ± Std. The best results are highlighted in **bold**. FD, 3D frequency decomposition; 3DSPE, 3D sinusoidal positional encoding; FKAN, Fourier KAN; PC, physics-inspired constraints loss.

Case	Components				Performance Indicators					
	L3D-WFDM		PC-FKAN		**Pavia University**			**Salinas**		
	FD	3DSPE	FKAN	PC	OA (%)	AA (%)	κ (%)	OA (%)	AA (%)	κ (%)
1	✓	×	×	×	83.57±0.20	70.98±0.35	77.68±0.30	91.64±0.43	83.58±1.09	90.66±0.49
2	✓	✓	×	×	84.23±0.72	71.68±1.86	78.57±1.00	92.21±0.51	84.20±1.09	91.29±0.57
4	×	×	✓	×	97.63±0.23	94.71±0.51	96.86±0.31	99.28±0.04	98.62±0.08	99.20±0.04
5	×	×	✓	✓	97.74±0.35	95.11±0.87	97.00±0.47	99.31±0.05	98.66±0.11	99.23±0.05
3	✓	×	✓	×	97.49±0.42	94.92±0.63	96.67±0.57	99.37±0.05	98.73±0.16	99.30±0.06
6	✓	×	✓	✓	97.54±0.18	94.91±0.45	96.74±0.24	99.40±0.07	98.78±0.16	99.33±0.08
7	✓	✓	✓	×	97.84±0.18	95.30±0.42	97.13±0.25	99.45±0.06	98.83±0.16	99.38±0.07
8 (FFKAN)	✓	✓	✓	✓	**98.54±0.03**	**96.91±0.23**	**98.06±0.03**	**99.60±0.05**	**99.25±0.15**	**99.55±0.06**
	L3D-WFDM		PC-FKAN		**Xuzhou**			**WHU-Hi-HanChuan**		
	FD	3DSPE	FKAN	PC	OA (%)	AA (%)	κ (%)	OA (%)	AA (%)	κ (%)
1	✓	×	×	×	91.10±0.42	83.72±0.75	88.60±0.54	82.84±0.65	65.35±1.21	79.73±0.76
2	✓	✓	×	×	91.41±0.46	84.22±1.09	89.00±0.60	83.73±0.61	66.65±0.81	80.77±0.71
4	×	×	✓	×	98.04±0.03	97.68±0.09	97.52±0.04	98.14±0.06	95.22±0.28	97.83±0.07
5	×	×	✓	✓	98.08±0.10	97.72±0.21	97.57±0.12	98.20±0.05	95.41±0.26	97.89±0.05
3	✓	×	✓	×	98.27±0.10	98.01±0.19	97.80±0.13	97.93±0.09	94.73±0.42	97.58±0.11
6	✓	×	✓	✓	98.39±0.06	98.18±0.12	97.95±0.08	98.01±0.11	95.32±0.31	97.67±0.13
7	✓	✓	✓	×	98.71±0.08	98.69±0.19	98.36±0.10	98.34±0.09	95.56±0.19	98.05±0.11
8 (FFKAN)	✓	✓	✓	✓	**98.95±0.03**	**98.79±0.10**	**98.67±0.03**	**98.50±0.06**	**95.89±0.13**	**98.24±0.07**

TABLE VIII: Performance metrics of different methods on the Pavia University, Salinas, Xuzhou, and WHU-Hi-HanChuan datasets

Dataset	Metric	SSFTT	LSGA	MASSFormer	S^2Mamba	SYNet	IGroupSS-Mamba	SliMamba	FFKAN
PU	Train(s)	5.33	7.48	4.96	**4.64**	11.81	40.36	7.71	6.71
	Test(s)	1.14	2.02	1.08	**0.87**	1.99	3.88	2.35	1.88
	FLOPs(M)	3.66	34.88	7.96	6.22	6.69	3.75	0.90	**0.32**
	Param(K)	148.03	422.29	303.85	76.65	162.56	57.31	13.32	**9.13**
SA	Train(s)	**5.75**	6.33	6.06	6.57	15.52	36.49	9.09	9.19
	Test(s)	1.21	1.62	1.27	**1.14**	2.56	4.71	4.14	2.36
	FLOPs(M)	3.66	34.88	7.96	6.75	6.69	3.75	0.97	**0.32**
	Param(K)	148.49	423.14	304.30	83.57	163.01	57.54	14.39	**9.36**
XZ	Train(s)	**7.61**	7.62	8.37	10.98	19.58	47.07	13.05	10.82
	Test(s)	1.74	2.07	1.75	**1.44**	3.41	5.94	7.71	2.90
	FLOPs(M)	3.66	34.88	7.96	7.95	6.69	3.75	1.12	**0.32**
	Param(K)	148.03	422.29	303.85	97.97	162.56	57.31	16.01	**9.13**
HC	Train(s)	**28.30**	30.51	28.86	30.55	72.06	266.58	42.57	40.36
	Test(s)	6.40	7.76	6.23	**5.38**	12.75	27.93	23.68	11.35
	FLOPs(M)	3.66	34.88	7.96	7.11	6.69	3.75	1.02	**0.32**
	Param(K)	148.49	423.14	304.30	88.05	163.01	57.54	14.96	**9.36**

evaluated under an identical patch size of 9×9 to ensure a fair benchmark.

As quantitatively presented in Table VIII, FFKAN achieves the lowest parameter count and theoretical FLOPs across all four HSI datasets. Requiring merely 9.13–9.36 K parameters and 0.32 M FLOPs, FFKAN drastically outperforms all competing methods in model compactness, where baselines range from 13.32 K to 423.14 K parameters and 0.90 M to 34.88 M FLOPs. This lightweight advantage stems from its tailored frequency-aware architecture: L3D-WFDM compresses spectral–spatial resolutions via separable 3D wavelet filtering, while PC-FKAN eliminates redundant feature transformations and limits Fourier modes. Consequently, FFKAN captures highly discriminative spectral–spatial features through effective frequency representation rather than parameter expansion.

Although FFKAN offers the smallest model size and computational load, its physical wall-clock runtime is not consistently the shortest. This discrepancy arises because GPU execution speed depends not only on floating-point arithmetic complexity, but also on memory bandwidth, operator types, and hardware execution efficiency. Specifically, multi-axis wavelet decomposition requires frequent tensor reshaping and transpositions, incurring negligible FLOPs but increasing memory access cost. Furthermore, the element-wise trigonometric functions and tensor contractions in Fourier KAN lack the hardware-level fused kernel acceleration enjoyed by standard convolutions. Auxiliary physical losses also add minor overhead exclusively during training. Nevertheless, FFKAN maintains highly competitive execution speeds, achieving an optimal balance among parameter efficiency, physical interpretability, and practical runtime.

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TABLE IX: OA (%) of different methods under the spatially disjoint evaluation protocol

Dataset	SSFTT	LSGA	MASSFormer	S^2Mamba	SYNet	IGroupSS-Mamba	SliMamba	FFKAN
PU	90.20±0.33	91.25±0.60	90.35±0.77	79.09±3.79	92.05±0.82	90.49±1.64	92.10±1.32	**93.14±0.43**
SA	84.68±0.79	84.98±1.27	84.09±1.39	76.06±3.52	82.21±1.24	84.64±3.77	84.47±2.04	**86.27±0.39**
XZ	95.52±0.33	95.05±0.61	96.07±0.41	93.09±0.44	94.67±0.45	95.08±0.74	94.58±0.86	**96.52±0.32**
HC	81.32±0.25	82.20±0.65	82.61±0.40	82.82±1.44	80.51±0.81	82.76±1.25	82.95±0.77	**83.54±0.80**

C. Spatially Disjoint Evaluation and Analysis

Patch-based HSI classification commonly uses random pixel sampling, where adjacent training and testing samples may produce overlapping patches and share local context, reducing their independence and potentially yielding optimistic results. To more rigorously assess spatial generalization and determine whether FFKAN benefits from such overlap, we further conduct a spatially disjoint evaluation.

Under this protocol, the image is divided into spatial blocks, and training samples of each class are selected from multiple blocks to improve spatial diversity. Candidate blocks are selected based on both class-specific sample quantity and their distance from previously selected blocks. Any candidate testing patch sharing at least one original image pixel with a training patch is excluded to ensure strict spatial separation. For all methods, approximately 1% of labeled samples per class are used for training with a fixed 9×9 patch. The block size is six times the patch size, with at least three blocks selected per class whenever possible. The weights for sample quantity and spatial distance are 0.6 and 0.4, respectively. Results are reported as mean $\pm$ standard deviation over five independent runs.

As shown in Table IX, all methods show performance degradation compared with random pixel sampling, with the largest decline observed on HC. Nevertheless, FFKAN achieves the best OA on all four datasets, indicating that its advantage does not primarily rely on shared local neighborhoods. The complementary frequency-aware modeling of L3D-WFDM and PC-FKAN may enable more transferable spectral–spatial representations under spatially separated conditions.

VI. Conclusion

This paper addresses the limited frequency-domain modeling capability of existing hyperspectral image classification methods by proposing FFKAN, a lightweight network for frequency-aware spectral–spatial representation learning. FFKAN comprises two principal modules. L3D-WFDM performs learnable 3D wavelet decomposition to adaptively disentangle low-frequency structural information from high-frequency discriminative details while retaining spectral–spatial positional cues. PC-FKAN subsequently models the decomposed features using learnable Fourier bases and incorporates physically inspired constraints to promote smooth, stable, and less redundant frequency representations. Experiments conducted on the Pavia University, Salinas, Xuzhou, and WHU-Hi-HanChuan datasets demonstrate that FFKAN achieves competitive or superior classification performance across diverse scenes while maintaining a compact parameter scale and low theoretical computational complexity. These results validate the effectiveness and lightweight nature of the proposed frequency-aware framework.

Despite these advantages, a certain discrepancy remains between the theoretical complexity of FFKAN and its practical inference latency. This discrepancy mainly arises from memory-access costs and operator-level overhead associated with sequential wavelet filtering, tensor rearrangement, and Fourier-layer computation. In addition, the generalization capability of FFKAN in larger-scale scenes and more diverse cross-sensor settings requires further evaluation. Future work will focus on developing more efficient wavelet and Fourier operators, reducing tensor-transformation overhead, and extending the framework to cross-scene, cross-sensor, and limited-label hyperspectral image classification.

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